

Project Expo — Problem Statement Bank

Problem Statement 01: Smart Tyre Pressure Warning System

Description:

Develop a system that detects abnormal tyre pressure and provides an early warning to the driver. The system should monitor pressure continuously or at defined intervals and indicate when the pressure moves outside a safe operating range. Students can demonstrate the concept using a real tyre, small pressure chamber, test setup, or simulated vehicle environment. This introduces sensing, signal processing, embedded electronics and vehicle safety without requiring a complete vehicle.

Reference:

The EU's current vehicle-safety framework includes tyre-pressure monitoring among the safety systems being introduced for road vehicles. Recent tyre recalls also continue to demonstrate that loss of tyre pressure can create safety risks; for example, NHTSA published a September 2025 Goodyear recall involving a condition that could cause air loss in service and increase crash risk.

[\(Reference Link\)](#)

Problem Statement 02: Brake-Light Failure Detection System

Description:

Develop a system that detects when a vehicle's brake light is not functioning correctly and alerts the driver or maintenance personnel. The prototype can use a miniature lighting circuit, current sensing or voltage monitoring to distinguish between normal and failed brake-light operation. The project teaches automotive electrical systems, fault detection, sensing and basic diagnostics.

Reference:

In September 2026, BYD recalled more than 180,000 vehicles because a brake-pedal component defect could cause the brake lights to remain illuminated when the pedal was not pressed, creating a safety risk for following drivers. This demonstrates the importance of monitoring brake-light behaviour rather than assuming the lighting system is always functioning correctly

[\(Reference Link\)](#)

Problem Statement 03: Low-Cost Brake Condition Monitoring System

Description:

Develop a prototype that identifies an abnormal brake condition using measurable parameters such as vibration, temperature, pressure, displacement or braking response. The system does not need to determine the exact remaining brake-pad life; instead, it should distinguish between

a normal and intentionally introduced abnormal condition and provide a warning. This gives students exposure to braking systems, sensors, mechanical behaviour and condition monitoring.

Reference:

Current vehicle inspection data continue to show brakes as a significant defect category. UK DVSA data for 2025–26 show brake-related defects among inspection failures, while the inspection guidance treats severely worn brake components as major or dangerous defects.

[\(Reference Link\)](#)

Problem Statement 04: Vehicle Cabin Air Quality Monitor

Description:

Develop a compact system that monitors vehicle cabin air quality and alerts the occupants when air quality deteriorates. Students can monitor parameters such as CO₂, particulate matter, temperature and humidity and create an indicator or warning system.

Reference:

Recent automotive research continues to investigate vehicle cabin air quality and pollutant exposure, particularly because vehicle occupants can be exposed to particulate matter and other pollutants during driving. Current automotive research is also moving toward real-time cabin monitoring and control.

[\(Reference Link\)](#)

Problem Statement 05: Driver Distraction Warning System

Description:

Develop a camera-based system that detects when a driver looks away from the road for an extended period and provides an appropriate warning. Students can demonstrate the system using a webcam and laptop rather than an actual moving vehicle. They should define measurable conditions such as head orientation or gaze-away duration and test the warning system under different situations.

Reference:

NHTSA's 2026 DrIIVE research specifically investigates drowsiness and distraction detection and mitigation. The EU is also implementing advanced driver-distraction warning requirements, making driver attention an active automotive safety-development area.

[\(Reference Link\)](#)

Problem Statement 06: Smart Part Quality Trend Monitor

Description:

Develop a simple digital system that monitors measurements of a manufactured automobile component and identifies when the process is beginning to move away from its acceptable range.

For example, students can create a small batch of identical components or use a supplied sample dataset containing measurements such as:

- hole diameter
- component thickness
- length
- weight
- hole position

The system should:

Collect measurements → compare with specification → display trend → identify abnormal variation → alert the operator.

The students don't need an industrial CNC machine or factory production line. They can use manual measurements + Excel/Python/web interface, or connect a digital measuring device if available.

[\(Reference Link\)](#)

Problem Statement 07: Intelligent Reverse Obstacle Warning System

Description:

Develop a system that detects obstacles behind a vehicle during reversing and provides a distance-dependent warning to the driver. The warning should become more urgent as the obstacle approaches. Students can build the prototype using ultrasonic sensing, camera-based detection or another suitable method and demonstrate it using a scaled vehicle model.

Reference:

Reversing detection using cameras or sensors is now included in the EU's mandatory vehicle-safety framework. Current automotive safety assessments also test systems against vulnerable road users and reversing scenarios.

[\(Reference Link\)](#)

Problem Statement 08: Camera-Based Speed-Limit Recognition System

Description:

Develop a system that identifies speed-limit signs from a road image or video and displays the detected speed limit to the driver. Students can begin with a small set of locally created road-sign images and build a computer-vision prototype. More advanced teams can investigate poor lighting, different sign angles or partially obscured signs.

Reference:

Current automotive safety systems increasingly integrate speed-limit information with camera inputs. Recent Euro NCAP assessments of 2025-model vehicles specifically evaluate how vehicles recognise speed limits using map and camera information.

[\(Reference Link\)](#)

Problem Statement 09: Pedestrian and Cyclist Collision Warning Prototype

Description:

Develop a prototype that detects a pedestrian or cyclist entering a vehicle's projected path and provides a warning to the driver. Students can use a camera, basic computer vision and a small model vehicle or video-based demonstration. The project should focus on detection and warning rather than attempting autonomous braking.

Reference:

Pedestrian and cyclist detection has become a major vehicle-safety-development area. Recent Euro NCAP testing includes automatic emergency braking scenarios involving pedestrians and cyclists, including crossing and roadside situations.

[\(Reference Link\)](#)

Problem Statement 10: Simple Vehicle Fault-Code Interpretation Assistant

Description:

Develop a small diagnostic assistant that converts vehicle fault information into a technician-friendly probable cause and inspection procedure.

For example:

Input:

Coolant temperature sensor fault

System output:

- Possible causes
- What should be checked first
- Relevant sensor/wiring
- Suggested diagnostic sequence

The prototype can be built as a desktop/web/mobile application using a limited set of predefined automotive diagnostic cases. It does not need to connect to an actual vehicle.

Reference:

SAE revised its worldwide On-Board Diagnostic requirements overview in July 2026, covering current OBD requirements across vehicle propulsion systems. SAE also revised J1939 diagnostic practices in September 2026, covering diagnostic trouble codes, ECU diagnostic information and vehicle parameter monitoring.

[\(Reference Link\)](#)

Problem Statement 11: EV Battery Over-Temperature Early Warning System

Description:

Develop a system that monitors battery temperature and provides an early warning when the temperature exceeds a defined safe threshold. Students can use a low-voltage battery pack or safe laboratory-scale cells and demonstrate controlled heating/temperature variation. The project should focus on sensing, threshold logic, warning and thermal monitoring—not intentionally creating thermal runaway.

Reference:

The Government of India reported in February 2026 that EV fire incidents remain an area of concern and described safety requirements covering batteries and BMS. Current EV testing requirements include over-temperature protection, over-current/short-circuit protection and other battery protections.

[\(Reference Link\)](#)

Problem Statement 12: Low-Cost EV Battery State-of-Health Estimator

Description:

Develop a prototype method to estimate the condition or remaining health of an EV battery using measurable parameters such as voltage, current, charge/discharge time and temperature. The prototype can use a small rechargeable battery rather than an actual automotive battery pack. Students should compare a healthy battery with a deliberately aged or lower-capacity battery and explain the limitations of their estimation method.

Reference:

A May 2026 peer-reviewed study specifically investigated the robustness of EV battery-pack state-of-health measurement and highlighted the influence of charging power, temperature, current direction and resting state of charge.

[\(Reference Link\)](#)

Problem Statement 13: EV Charging Fault Detection System

Description:

Develop a system that identifies common charging abnormalities such as loss of connection, abnormal current, communication failure or charging interruption and provides a clear diagnostic indication. Students can build a low-voltage charging demonstrator rather than working with high-voltage EV charging equipment. The objective is to teach fault detection, electrical measurement and charging-system logic.

Reference:

India's EV charging network has expanded rapidly, while recent industry analysis identifies connector wear, software/protocol mismatches, payment failures, connectivity and environmental conditions among causes of charging-system downtime. Government data also show more than 29,000 public EV charging stations in India.

[\(Reference Link\)](#)

Problem Statement 14: Vehicle Auxiliary Power Cut-Off Safety System

Description:

Develop a system that automatically disconnects a vehicle's auxiliary electrical load when an abnormal electrical condition is detected.

The prototype could monitor:

- excessive current
- prolonged operation
- abnormal temperature
- stalled motor/load

and automatically disconnect the affected load while giving a warning.

Reference:

NHTSA issued a June 2026 recall covering more than 1 million Jeep Wrangler and Gladiator vehicles because an electrical connection problem in the electric hydraulic power-steering pump wiring could overheat and cause a fire. NHTSA reported 51 fires associated with the issue.

[\(Reference Link\)](#)

Problem Statement 15: Smart EV Charging Load Management System

Description:

Develop a system that schedules EV charging based on available electrical capacity, charging priority and time. The prototype could simulate several EVs requesting charging simultaneously and decide when each vehicle should charge to reduce peak demand. Students can implement the system using a microcontroller, software simulation or both.

Reference:

The IEA's 2026 Global EV Outlook highlights rapidly growing charging demand and identifies smart charging as a method of shifting charging loads to provide flexibility. India is also rapidly expanding its charging infrastructure.

[\(Reference Link\)](#)

Problem Statement 16: EV Battery Second-Life Screening System

Description:

Develop a simple method to determine whether a used EV battery module is suitable for further low-power applications such as stationary energy storage. Students can create a safe low-voltage demonstration using rechargeable cells and assess capacity, voltage consistency and basic health indicators. The project should focus on screening and decision-making, not dismantling high-voltage EV packs.

Reference:

Recent Indian research and policy work identifies end-of-life EV batteries as an emerging circular-economy challenge. A 2026 NITI Aayog document reports that India has significant gaps in formal collection and recycling infrastructure for lithium-ion batteries, while current recovery targets are increasing.

[\(Reference Link\)](#)

Problem Statement 17: AI-Based Automotive Component Defect Detection

Description:

Develop a computer-vision system that identifies visible defects in a selected automobile component—for example, scratches, missing features, incorrect orientation, holes or surface defects. Students can create their own small dataset using manufactured/sample parts and train or configure a simple computer-vision model. The emphasis should be on comparing detection accuracy and explaining false positives/false negatives rather than merely displaying an AI model.

Reference:

This is directly connected to current automotive manufacturing practice. A 2026 SAE technical paper from Tata Motors researchers describes AI and CAD-data-based defect detection for automotive manufacturing, while a 2025 open-access study demonstrated computer-vision inspection of automotive components.

[\(Reference Link\)](#)

Problem Statement 18: Secure Automotive Software Update Verification Prototype

Description:

Develop a small software prototype that demonstrates how an automotive electronic control system could verify whether a software update is genuine and compatible before accepting it. Students can use a simulated ECU environment rather than an actual vehicle. The system could demonstrate version checking, file integrity verification and rejection of an altered or incompatible update. This introduces students to automotive cybersecurity and software-defined vehicles.

Reference:

Automotive software and cybersecurity remain active engineering and regulatory areas. UNECE's 2025–26 working groups continued developing requirements and interpretations around vehicle software updates, including UN Regulation No. 156. The IEA's 2026 Global EV Outlook also identifies increasingly software-defined vehicles and growing cybersecurity importance.

[\(Reference Link\)](#)
